

PRACTICE SHEET

1. If $\int \sec x \operatorname{cosec} x \, dx = \log|g(x)| + c$, then what is $g(x)$ equal to?
 (a) $\sin x \cos x$ (b) $\sec^2 x$
 (c) $\tan x$ (d) $\log|\tan x|$
2. What is the value of $\int (\sqrt{x} + x)^{-1} \, dx$?
 (a) $\ln(x + \sqrt{x}) + c$ (b) $2\ln(1 + \sqrt{x}) + c$
 (c) $2\ln(x + \sqrt{x}) + c$ (d) $2\ln(x - \sqrt{x}) + c$
3. What is the value of $\int \frac{e^x(1+x)}{\sin^2(xe^x)} \, dx$?
 (a) $-e^x \cot x + c$ (b) $\cos 2(xe^x) + c$
 (c) $\log \sin(xe^x) + c$ (d) $-\cot(xe^x) + c$
4. What is $\int \log(x+1) \, dx$ is equal to?
 (a) $x \log(x+1) - x + c$
 (b) $(x+1) \log(x+1) - x + c$
 (c) $\frac{1}{x+1} + c$
 (d) $\frac{\log(x+1)}{x+1} + c$
5. If $\int \frac{dx}{f(x)} = \log\{f(x)\}^2 + c$, then what is (x) equal to?
 (a) $2x + \alpha$ (b) $x + \alpha$
 (c) $\frac{x}{2} + \alpha$ (d) $x^2 + \alpha$
6. What is $\int (e^x + 1)^{-1} \, dx$ equal to?
 (a) $\ln(e^x + 1) + c$ (b) $\ln(e^{-x} + 1) + c$
 (c) $-\ln(e^{-x} + 1) + c$ (d) $-(e^x + 1) + c$
7. What is $\int \frac{d\theta}{\sin^2 \theta + 2 \cos^2 \theta - 1}$ equal to?
 (a) $\tan \theta + c$ (b) $\cot \theta + c$
 (c) $\frac{1}{2} \tan \theta + c$ (d) $\frac{1}{2} \cot \theta + c$
8. What is $\int \sin x \log(\tan x) \, dx$ equal to?
 (a) $\cos x \log \tan x + \log \tan(x/2) + c$
 (b) $-\cos x \log \tan x + \log \tan(x/2) + c$
 (c) $\cos x \log \tan x + \log \cot(x/2) + c$
 (d) $-\cos x \log \tan x + \log \cot(x/2) + c$
9. What is $\int \tan^2 x \sec^4 x \, dx$ equal to?
 (a) $\frac{\sec^5 x}{5} + \frac{\sec^3 x}{3} + c$ (b) $\frac{\tan^5 x}{5} + \frac{\tan^3 x}{3} + c$
- (c) $\frac{\tan^5 x}{5} + \frac{\sec^3 x}{3} + c$ (d) $\frac{\sec^5 x}{5} + \frac{\tan^3 x}{3} + c$
10. What $\int \sec x^\circ \, dx$ is equal to?
 (a) $\log(\sec x^\circ + \tan x^\circ) + c$
 (b) $\frac{\pi \log \tan\left(\frac{\pi}{4} + \frac{\pi}{2}\right)}{180^\circ} + c$
 (c) $\frac{180^\circ \log \tan\left(\frac{\pi}{4} + \frac{\pi}{2}\right)}{\pi} + c$
 (d) $\frac{180^\circ \log \tan\left(\frac{\pi}{4} + \frac{x}{360^\circ}\right)}{\pi} + c$
11. What is $\int \frac{a + b \sin x}{\cos^2 x} \, dx$ equal to?
 (a) $a \sec x + b \tan x + c$
 (b) $a \tan x + b \sec x + c$
 (c) $a \cot x + b \operatorname{cosec} x + c$
 (d) $a \operatorname{cosec} x + b \cot x + c$
12. What is $\int \frac{\log x}{(1 + \log x)^2} \, dx$ equal to?
 (a) $\frac{1}{(1 + \log x)^3} + c$ (b) $\frac{1}{(1 + \log x)^2} + c$
 (c) $\frac{x}{(1 + \log x)} + c$ (d) $\frac{x}{(1 + \log x)^2} + c$
13. What is $\int \sec^n x \tan x \, dx$ equal to?
 (a) $\frac{\sec^n x}{n} + C$ (b) $\frac{\sec^{n-1} x}{n-1} + C$
 (c) $\frac{\tan^n x}{n} + C$ (d) $\frac{\tan^{n-1} x}{n-1} + C$
14. What is $\int (e^{-x} + 1)^{-1} \, dx$ equal to?
 (a) $\log(e^x + 1) + C$ (b) $\log(e^x - 1) + C$
 (c) $\log(e^{-x} + 1) + C$ (d) $\log(e^{-x} - 1) + C$
15. What is $\int \frac{x^4 + 1}{x^2 + 1} \, dx$ equal to?
 (a) $\frac{x^3}{3} - x + 4 \tan^{-1} x + C$ (b) $\frac{x^3}{3} + x + 4 \tan^{-1} x + C$
 (c) $\frac{x^3}{3} - x + 2 \tan^{-1} x + C$ (d) $\frac{x^3}{3} - x - \tan^{-1} x + C$

ANSWER KEY

1.	c	2.	b	3.	d	4.	b	5.	c	6.	c	7.	a	8.	b	9.	b	10.	a
11.	b	12.	c	13.	a	14.	a	15.	c										

Solutions

Sol. 1. (c)

$$\text{Let } I = \int \sec x \cdot \operatorname{cosec} x \, dx = \int \frac{2}{2 \sin x \cos x} \, dx$$

$$= 2 \int \frac{1}{\sin 2x} \, dx = 2 \int \frac{1}{2 \tan x} \frac{1}{1 + \tan^2 x} \, dx$$

$$\left[\because \sin 2x = \frac{2 \tan x}{1 + \tan^2 x} \right]$$

$$= \int \frac{\sec^2 x}{\tan x} \, dx$$

$$\text{Let } \tan x = t \Rightarrow \sec^2 x \, dx = dt$$

$$\text{So, } I = \int \frac{dt}{t} \log |t| + c = \log |\tan x| + c$$

$$g(s) = \tan x$$

$$\text{But } \int \sec x \operatorname{cosec} x \, dx = \log |g(x)| + c$$

$$\therefore g(x) = \tan x$$

Sol. 2. (b)

$$\int (x + \sqrt{x})^{-1} \, dx = \int \left(\frac{1}{x + \sqrt{x}} \right) dx$$

$$= \int \frac{1}{\sqrt{x} \cdot \sqrt{x} + \sqrt{x}} \, dx$$

$$\text{Let } \sqrt{x} + 1 = t$$

$$\text{then, } \frac{1}{2\sqrt{x}} \, dx = dt \Rightarrow \frac{1}{\sqrt{x}} \, dx = 2dt$$

$$\therefore \int \frac{1}{\sqrt{x}(\sqrt{x} + 1)} \, dx = \int \frac{2dt}{t} = 2 \log t + c$$

$$= 2 \log (\sqrt{x} + 1) + c$$

Sol. 3. (d)

Let the given integral be

$$I = \int \frac{e^x(1+x)}{\sin^2(xe^x)} \, dx$$

$$\text{Put } x e^x = t \text{ and } e^x(1+x)dx = dt$$

$$\Rightarrow I = \int \frac{dt}{\sin^2 t} = \int \operatorname{cosec}^2 t \, dt$$

$$= -\cot t + c = -\cot(xe^x) + c$$

Sol. 4. (b)

$$\text{Let } I = \int \log(x+1) \, ds$$

$$\text{Let } x+1 = t$$

$$\Rightarrow dx = dt$$

$$\Rightarrow I = \int 1 \cdot \log t \, dt$$

Integration by parts, taking $\log t$ as first function

$$\Rightarrow I = t \log t - \int 1 \cdot t \, dt + c_1$$

$$\Rightarrow t \log t - \int 1 \, dt + c_1 = t \log t - t + c_1$$

$$= (x+1) \log(x+1) - x - 1 + c_1$$

$$= (x+1) \log(x+1) - x + c \quad [\because c = c_1 - 1]$$

Sol. 5. (c)

We check from the given options one by one. Options (a) and (b) do not satisfy.

We check option (c)

$$\text{Let } f(x) = x/2 + \alpha$$

$$\therefore \int \frac{dx}{\frac{x}{2} + \alpha} = \int \frac{dx}{(x+2\alpha)}$$

$$= 2 \log(x+2\alpha) + c_1 = \log(x+2\alpha) + c_1$$

$$= \log\left(\frac{x}{2} + \alpha\right)^2 + \log 2^2 + c_1 = \log\left(\frac{x}{2} + \alpha\right)^2 + c$$

Sol. 6. (c)

$$\text{Let } I = \int (e^x + 1)^{-1} \, dx = \int \frac{1}{e^x + 1} = \int \frac{e^{-x}}{1 + e^{-x}} \, dx$$

$$\text{Let } 1 + e^{-x} = t \Rightarrow -e^{-x} \, dx = dt$$

$$\therefore I = -\int 1/t \, dt = -\log t + c = -\log(1 + e^{-x}) + c$$

Sol. 7. (a)

$$\text{Let } I = \int \frac{d\theta}{\sin^2 \theta + 2 \cos^2 \theta - 1}$$

$$= \int \frac{d\theta}{1 - \cos^2 \theta + 2 \cos^2 \theta - 1} = \int \frac{d\theta}{\cos^2 \theta}$$

$$= \int \sec^2 \theta \, d\theta = \tan \theta + c$$

Sol. 8. (b)

$$\int \sin x \log(\tan x) \, dx$$

$$= -\cos x \log \tan x - \int (-\cos x) \cdot \frac{1}{\sin x} \cdot \sec^2 x \, dx$$

$$= -\cos x \log \tan x + \int \frac{1}{\sin x} \, dx$$

$$= -\cos x \log(\tan x) + \int \frac{1 + \tan^2 x}{2 \tan x} \, dx$$

$$\text{Let } t = \tan x/2$$

$$\Rightarrow \frac{dx}{dt} = \frac{2}{1+t^2} \Rightarrow dx = \frac{2}{1+t^2} \cdot dt$$

$$\text{So, } -\cos x \cdot \log(\tan x) + \int \frac{1 + \tan^2 x}{2 \tan x} \, dx$$

$$= -\cos x \cdot \log(\tan x) + \int \frac{1+t^2}{2t} \cdot \frac{2}{1+t^2} \, dt$$

$$= -\cos x \log \tan x + \log(t) + c$$

$$= -\cos x \log \tan x + \log \tan\left(\frac{x}{2}\right) + c$$

Sol. 9. (b)

$$\text{Let } I = \int \tan^2 x \sec^4 x \, dx$$

$$\text{Let } \tan x = t$$

$$\Rightarrow \sec^2 x \, dx = dt$$

$$\therefore I = \int \tan^2 x \cdot \sec^2 x \cdot \sec^2 x \cdot dx$$

$$= \int \tan^2 x (1 + \tan^2 x) \sec^2 x \cdot dx$$

$$\therefore I = \int t^2 (1+t^2) \, dt = \int (t^2 + t^4) \, dt$$

$$= \frac{t^2}{5} + \frac{t^2}{3} + c = \frac{\tan^5 x}{5} + \frac{\tan^3 x}{3} + c$$

Sol. 10. (a)

$$\int \sec x^\circ \cdot dx = \int \frac{\sec x^\circ (\sec x^\circ + \tan x^\circ)}{\sec x^\circ + \tan x^\circ} \cdot dx$$

$$\text{Let } u = \sec x^\circ + \tan x^\circ$$

$$\Rightarrow \frac{du}{dx} = \sec x^\circ + \tan x^\circ + \sec 2x^\circ$$

$$\Rightarrow du = (\sec x^\circ + \tan x^\circ + \sec 2x^\circ) \cdot dx$$

$$\therefore \int \frac{\sec x^\circ (\sec x^\circ + \tan x^\circ)}{\sec x^\circ + \tan x^\circ} \cdot dx$$

$$= \int \frac{\sec^2 x^\circ + \sec x^\circ \cdot \tan x^\circ}{\sec x^\circ + \tan x^\circ} \cdot dx$$

$$= \int \frac{du}{u} = \log(u) + C = \log(\sec x^\circ + \tan x^\circ) + C$$

Sol. 11. (b)

$$\text{Consider } \int \frac{a + b \sin x}{\cos^2 x} \, dx = \int \left(\frac{a}{\cos^2 x} + \frac{b \sin x}{\cos^2 x} \right) dx$$

$$= \int (a \sec^2 x + b \tan x \sec x) \, dx = a \tan x + b \sec x + c$$

Sol. 12. (c)

$$\text{Let } I = \int \frac{\log x}{(1 + \log x)^2} \, dx$$

$$\text{Put } \log x = t \Rightarrow 1/x \, dx = dt$$

$$\Rightarrow dx = x \cdot dt \Rightarrow dx = e^t \cdot dt \quad (\because x = e^t)$$

$$I = \int \frac{e^t t}{(1+t)^2} \, dt = \int \frac{e^t \cdot (t+1-1)}{(1+t)^2} \, dt$$

$$= \int \frac{e^t(1+t)}{(1+t)^2} \, dt - \int \frac{e^t}{(1+t)^2} \, dt = \int \frac{e^t}{1+t} \, dt - \int \frac{e^t}{(1+t)^2} \, dt$$

$$= \frac{e^t}{1+t} - \int -e^t \cdot \frac{1}{(1+t)^2} \, dt - \int \frac{e^t}{(1+t)^2} \, dt$$

$$= \frac{e^t}{1+t} + \int e^t \cdot \frac{1}{(1+t)^2} \, dt - \int \frac{e^t}{(1+t)^2} \, dt$$

$$= \frac{e^t}{1+t} + c = \frac{x}{1 + \log x} + c$$

Sol. 13. (a)

$$\int \sec^n x \tan x \, dx = \int \sec^{n-1} x \cdot \sec x \cdot \tan x \, dx,$$

$$\text{put } \sec x = t \text{ then } \tan x \cdot \sec x \, dx = dt$$

$$= \int t^{n-1} \, dt = t^n/n + c$$

$$= \frac{\sec^n x}{n} + C$$

Sol. 14. (a)

$$\int (e^{-x} + 1)^{-1} \, dx = \int \frac{dx}{(e^{-x} + 1)}$$

$$= \int \frac{e^x \, dx}{(1 + e^x)} = \log(1 + e^x) + c$$

Sol. 15. (c)

$$\int \frac{x^x + 1}{x^2 + 1} \, dx = \int \frac{x^4 - 1 + 2}{x^2 + 1} \, dx$$

$$\int \frac{x^4 - 1}{x^2 + 1} \, dx + \int \frac{2}{x^2 + 1} \, dx$$

$$= \int (x^2 - 1) \, dx + \int \frac{2}{x^2 + 1} \, dx$$

$$= x^3/3 - x + 2 \tan^{-1} x + C$$

SUBJECTIVE PRACTICE SHEET

Question based on formulae:

1. $\int \frac{x^3 - 1}{x^2} dx$
2. $\int \cos ecx (\cos ecx + \cot x) dx$
3. $\int \frac{1 - \sin x}{\cos^2 x} dx$
4. $\int \left(\sqrt{x} - \frac{1}{\sqrt{x}} \right)^2 dx$
5. $\int \frac{x^3 - x^2 + x - 1}{x - 1} dx$
6. $\int (1 - x) \sqrt{x} dx$
7. $\int secx (secx + \tan x) dx$
8. $\int \frac{\sec^2 x}{\cos ec^2 x} dx$
9. $\int \frac{1 - 3 \sin x}{\cos^2 x} dx$
10. $\int \cos^2 x dx$
11. $\int \sin^2 (2x + 5) dx$
12. $\int \sin 3x \cos 4x dx$
13. $\int \cos 2x \cos 4x \cos 6x dx$
14. $\int \frac{1 - \cos x}{1 + \cos x} dx$
15. $\int \frac{\cos x}{1 + \cos x} dx$
16. $\int \frac{\sin^2 x}{1 + \cos x} dx$
17. $\int \sin^{-1} (\cos x) dx$
18. $\int \sin 2x \cos 3x dx$
19. $\int \frac{1}{\sin^2 x \cos^2 x} dx$

Questions based on Substitution

20. $\int \sin^3 x \cos^2 x dx$
21. $\int \frac{2x}{1 + x^2} dx$
22. $\int \frac{(\log x)^2}{x} dx$
23. $\int \frac{1}{x + x \log x} dx$
24. $\int \sin x \sin (\cos x) dx$
25. $\int \sqrt{ax + b} dx$
26. $\int x \sqrt{x + 2} dx$
27. $\int x \sqrt{1 + 2x^2} dx$
28. $\int (4x + 2) \sqrt{x^2 + x + 1} dx$
29. $\int \frac{1}{x - \sqrt{x}} dx$
30. $\int \frac{x}{\sqrt{x + 4}} dx$
31. $\int (x^3 - 1)^{1/3} x^5 dx$
32. $\int \frac{x}{9 - 4x^2} dx$
33. $\int \frac{x}{e^{x^2}} dx$
34. $\int \frac{e^{\tan^{-1} x}}{1 + x^2} dx$
35. $\int \frac{e^{2x} - 1}{e^{2x} + 1} dx$
36. $\int \frac{e^{2x} - e^{-2x}}{e^{2x} + e^{-2x}} dx$
37. $\int \frac{1}{1 + \tan x} dx$
38. $\int \frac{1}{1 + \cot x} dx$

39. $\int \frac{\cos \sqrt{x}}{\sqrt{x}} dx$
40. $\int \cot x \log \sin x dx$
41. $\int \frac{1}{\cos^2 x (1 - \tan x)^2} dx$
42. $\int \frac{\cos x}{\sqrt{1 + \sin x}} dx$
43. $\int \frac{\sqrt{\tan x}}{\sin x \cos x} dx$
44. $\int \sec^4 x dx$
45. $\int \sin^3 x dx$
46. $\int \frac{1}{\sin x \cos^3 x} dx$
47. $\int \frac{\cos x - \sin x}{1 + \sin 2x} dx$
48. $\int \frac{(x + 1)(x + \log x)^2}{x} dx$
49. $\int \frac{x^3 \sin (\tan^{-1} x^4)}{1 + x^8} dx$

Integration by Substitutional formulae:

50. $\int \frac{3x^2}{x^6 + 1} dx$
51. $\int \frac{1}{\sqrt{1 + 4x^2}} dx$
52. $\int \frac{3x}{1 + 2x^4} dx$
53. $\int \frac{x - 1}{\sqrt{x^2 - 1}} dx$
54. $\int \frac{x^2}{\sqrt{x^6 + a^6}} dx$
55. $\int \frac{\sec^2 x}{\sqrt{\tan^6 x + 4}} dx$

56. $\int \frac{1}{\sqrt{x^2 + 2x + 2}} dx$

57. $\int \frac{1}{\sqrt{7 - 6x - x^2}} dx$

58. $\int \frac{1}{\sqrt{(x-1)(x-2)}} dx$

59. $\int \frac{1}{\sqrt{8 + 3x - x^2}} dx$

60. $\int \frac{4x+1}{\sqrt{2x^2 + x - 3}} dx$

61. $\int \frac{x+2}{\sqrt{x^2 - 1}} dx$

62. $\int \frac{x+2}{\sqrt{x^2 + 2x + 3}} dx$

63. $\int \frac{5x+3}{\sqrt{x^2 + 4x + 10}} dx$

64. $\int \sqrt{4 - x^2} dx$

65. $\int \sqrt{1 - 4x^2} dx$

66. $\int \sqrt{x^2 + 4x + 6} dx$

67. $\int \sqrt{x^2 + 4x + 1} dx$

68. $\int \sqrt{1 - 3x - x^2} dx$

69. $\int \sqrt{1 + \frac{x^2}{9}} dx$

Integration by Parts:

70. $\int x \sin x dx$

71. $\int x \sin 3x dx$

72. $\int x^2 e^x dx$

73. $\int x \log x dx$

74. $\int x \log 2x dx$

75. $\int x^2 \log x dx$

76. $\int x \sin^{-1} x dx$

77. $\int x \tan^{-1} x dx$

78. $\int (\sin^{-1} x)^2 dx$

79. $\int x (\log x)^2 dx$

80. $\int e^x (\sin x + \cos x) dx$

81. $\int \frac{xe^x}{(1+x)^2} dx$

82. $\int e^x \left(\frac{1 + \sin x}{1 + \cos x} \right) dx$

83. $\int e^x \left(\frac{1}{x} - \frac{1}{x^2} \right) dx$

84. $\int \frac{(x-3)e^x}{(1-x)^3} dx$

85. $\int e^x \sec x (1 + \tan x) dx$

86. $\int e^x \left(\tan^{-1} x + \frac{1}{1+x^2} \right) dx$

87. $\int \frac{(x^2+1)e^x}{(x+1)^2} dx$

88. $\int e^x \sin x dx$

89. $\int e^{2x} \sin x dx$

Integration by partial fraction:

90. $\int \frac{1}{x^2 - 9} dx$

91. $\int \frac{x}{(x+1)(x+2)} dx$

92. $\int \frac{2x}{x^2 + 3x + 2} dx$

93. $\int \frac{1}{x(x^n + 1)} dx$

94. $\int \frac{1}{x(x^2 + 1)} dx$

Answers

Ans. 1. $\frac{x^2}{2} + \frac{1}{x} + c$

Ans. 2. $-\cot x - \cos ecx + c$

Ans. 3. $\tan x - secx + c$

Ans. 4. $\frac{x^2}{2} - 2x + \log x + c$

Ans. 5. $\frac{x^3}{3} + x + c$

Ans. 6. $\frac{2}{3}x^{3/2} - \frac{2}{5}x^{5/2} + c$

Ans. 7. $\tan x + secx + c$

Ans. 8. $\tan x - x + c$

Ans. 9. $\tan x - 3secx + c$

Ans. 10. $\frac{x}{2} + \frac{1}{4} \sin 2x + c$

Ans. 11. $\frac{x}{2} - \frac{1}{8} \sin(4x + 10) + c$

Ans. 12. $-\frac{1}{14} \cos 7x + \frac{1}{2} \cos x + c$

Ans. 13. $\frac{1}{4} \left[\frac{1}{12} \sin 12x + x + \frac{1}{8} \sin 8x + \frac{1}{4} \sin 4x \right] + c$

Ans. 14. $2 \tan \frac{x}{2} - x + c$

Ans. 15. $x - \tan \frac{x}{2} + c$

Ans. 16. $x - \sin x + c$

Ans. 17. $\frac{\pi x}{2} - \frac{x^2}{2} + c$

Ans. 18. $-\frac{1}{10} \cos 5x + \frac{1}{2} \cos x + c$

Ans. 19. $\tan x - \cot x + c$

Ans. 20. $-\frac{1}{3} \cos^3 x + \frac{1}{5} \cos^5 x + c$

Ans. 21. $\log(1+x^2) + c$

Ans. 22. $\frac{1}{3} (\log x)^3 + c$

Ans. 23. $\log(1+\log x) + c$

Ans. 24. $\cos(\cos x) + c$

Ans. 25. $\frac{2}{3a} (ax+b)^{\frac{3}{2}} + c$

Ans. 26. $\frac{2}{5} (x+2)^{\frac{5}{2}} - \frac{4}{3} (x+2)^{\frac{3}{2}} + c$

Ans. 27. $\frac{1}{6} (1+2x^2)^{\frac{3}{2}} + c$

Ans. 28. $\frac{4}{3} (x^2+x+1)^{\frac{3}{2}} + c$

Ans. 29. $2 \log |\sqrt{x}-1| + c$

Ans. 30. $\frac{2}{3} \sqrt{x+4} (x-8) + c$

Ans. 31. $\frac{1}{7} (x^3-1)^{\frac{7}{3}} + \frac{1}{4} (x^3-1)^{\frac{4}{3}} + c$

Ans. 32. $-\frac{1}{8} \log |9-4x^2| + c$

Ans. 33. $-\frac{1}{2e^{x^2}} + c$

Ans. 34. $e^{\tan^{-1} x} + c$

Ans. 35. $\log(e^x + e^{-x}) + c$

Ans. 36. $\frac{1}{2} \log(e^{2x} + e^{-2x}) + c$

Ans. 37. $\frac{x}{2} + \log(\cos x + \sin x) + c$

Ans. 38. $\frac{x}{2} - \frac{1}{2} \log(\cos x + \sin x) + c$

Ans. 39. $2 \sin \sqrt{x} + c$

Ans. 40. $\frac{(\log \sin x)^2}{2} + c$

Ans. 41. $\frac{1}{1-\tan x} + c$

Ans. 42. $2\sqrt{1+\sin x} + c$

Ans. 43. $2\sqrt{\tan x} + c$

Ans. 44. $\tan x + \frac{\tan^3 x}{3} + c$

Ans. 45. $-\cos x + \frac{1}{3} \cos^3 x + c$

Ans. 46. $\log \tan x + \frac{1}{2} \tan^2 x + c$

Ans. 47. $-\frac{1}{\cos x + \sin x} + c$

Ans. 48. $\frac{1}{3} (x + \log x)^3 + c$

Ans. 49. $-\frac{1}{4} \cos(\tan^{-1} x^4) + c$

Ans. 50. $\tan^{-1} x^3 + c$

Ans. 51. $\frac{1}{2} \log(2x + \sqrt{1+4x^2})$

Ans. 52. $\frac{3}{2\sqrt{2} \tan^{-1} \sqrt{2} x^2} + c$

Ans. 53. $\sqrt{x^2-1} - \log(x + \sqrt{x^2-1}) + c$

Ans. 54. $\frac{1}{3} \log(x^3 + \sqrt{x^6+a^6}) + c$

Ans. 55. $\log(\tan x + \sqrt{\tan^2 x + 4}) + c$

Ans. 56. $\log(x+1 + \sqrt{x^2+2x+2}) + c$

Ans. 57. $\sin^{-1}\left(\frac{x+3}{4}\right)$

Ans. 58. $\log\left(x - \frac{3}{2} + \sqrt{x^2-3x+2}\right) + c$

Ans. 59. $\sin^{-1}\left(\frac{x+3}{4}\right)$

Ans. 60. $2\sqrt{2x^2+x-3} + c$

Ans. 61. $\sqrt{x^2 - 1} + 2 \log(x + \sqrt{x^2 - 1}) + c$

Ans. 62. $\sqrt{x^2 + 2x + 3} + \log(x + 1 + \sqrt{x^2 + 2x + 3}) + c$

Ans. 63. $5\sqrt{x^2 + 4x + 10} - 7 \log(x + 2 + \sqrt{x^2 + 4x + 10}) + c$

Ans. 64. $\frac{1}{2}x\sqrt{4 - x^2} + 2 \sin^{-1} \frac{x}{2} + c$

Ans. 65. $\frac{1}{4} \sin^{-1} 2x + \frac{1}{2}x\sqrt{1 - 4x^2} + c$

Ans. 66. $\frac{x+2}{2}\sqrt{x^2 + 4x + 6} + \log(x + 2 + \sqrt{x^2 + 4x + 6}) + c$

Ans. 67. $\frac{x+2}{2}\sqrt{x^2 + 4x + 1} - \frac{3}{2} \log(x + 2 + \sqrt{x^2 + 4x + 1}) + c$

Ans. 68. $\frac{2x-3}{4}\sqrt{1+3x-x^2} + \frac{13}{8} \sin^{-1}\left(\frac{2x-13}{\sqrt{13}}\right) + c$

Ans. 69. $\frac{x}{6}\sqrt{x^2 + 9} + \frac{3}{2} \log(x + \sqrt{x^2 + 9}) + c$

Ans. 70. $-x \cos x + \sin x + c$

Ans. 71. $-\frac{x}{3} \cos 3x + \frac{1}{9} \sin 3x + c$

Ans. 72. $e^x(x^2 - 2x + 2) + c$

Ans. 73. $\frac{x^2}{2} \log x - \frac{x^2}{4} + c$

Ans. 74. $\frac{x^2}{2} \log 2x - \frac{x^2}{4} + c$

Ans. 75. $\frac{x^2}{3} \log x - \frac{x^3}{9} + c$

Ans. 76. $(2x^2 - 1) \frac{\sin^{-1} x}{4} + \frac{x\sqrt{1-x^2}}{4} + c$

Ans. 77. $\frac{x^2}{2} \tan^{-1} x - \frac{x}{2} + \frac{1}{2} \tan^{-1} x + c$

Ans. 78. $(\sin^{-1} x)^2 + 2\sqrt{1-x^2} \sin^{-1} x - 2x + c$

Ans. 79. $\frac{x^2}{2} (\log x)^2 - \frac{x^2}{2} \log x + \frac{x^2}{4} + c$

Ans. 80. $e^x \sin x + c$

Ans. 81. $e^x \cdot \frac{1}{1+x^2} + c$

Ans. 82. $e^x \tan \frac{x}{2} + c$

Ans. 83. $\frac{e^x}{x} + c$

Ans. 84. $\frac{e^x}{(x-1)^2} + c$

Ans. 85. $e^x \sec x + c$

Ans. 86. $e^x \tan^{-1} x + c$

Ans. 87. $e^x \frac{x-1}{x+1} + c$

Ans. 88. $\frac{e^x}{2} (\sin x - \cos x) + c$

Ans. 89. $\frac{e^{2x}}{5} (2 \sin x - \cos x) + c$

Ans. 90. $\frac{1}{6} \log \left(\frac{x-3}{x+3} \right)$

Ans. 91. $\log \frac{(x+2)^2}{x+1} + c$

Ans. 92. $4 \log(x+2) - 2 \log(x+1) + c$

Ans. 93. $\frac{1}{n} \log \frac{x^n}{x^n + 1} + c$

Ans. 94. $\frac{1}{2} \log \frac{x^2}{x^2 + 1} + c$

NDA PYQ

1. What is $\int \frac{\sin \sqrt{x}}{\sqrt{x}} dx$ is equal to?
 (a) $\frac{\cos \sqrt{x}}{2} + C$ (b) $2 \cos \sqrt{x} + C$
 (c) $\frac{-\cos \sqrt{x}}{2} + C$ (d) $-2 \cos \sqrt{x} + C$
[NDA (I) - 2011]
 2. What is the value of $\int e^x \left(\sqrt{x} + \frac{1}{2\sqrt{x}} \right) dx$?
 (a) $xe^x + c$ (b) $e^x \sqrt{x} + c$
 (c) $2e^x \sqrt{x}$ (d) $2xe^x + c$
[NDA (I) - 2011]
 3. What is $\int \sin^{-1}(\cos x) dx$ equal to?
 (a) $\frac{x\pi}{2} - \frac{x^2}{2} + k$ (b) $\frac{\pi}{2} + \frac{x^2}{2} + k$
 (c) $-\frac{x\pi}{2} - \frac{x^2}{2} + k$ (d) $\frac{\pi}{2} - \frac{\pi^2}{2} + k$
[NDA (I) - 2011]
 4. What is $\int \frac{dx}{\sin^2 x \cos^2 x}$ equal to?
 (a) $\tan x + \cot x + c$
 (b) $\tan x - \cot x + c$
 (c) $(\tan x + \cot x)^2 + c$
 (d) $(\tan x - \cot x)^2 + c$
[NDA-2011(2)]
 5. What is the value of $\int (x^2+1)^{5/2} x dx$?
 (a) $(x^2+1)^{7/2} + C$ (b) $\frac{2}{7}(x^2+1)^{7/2} + C$
 (c) $\frac{1}{7}(x^2+1)^{7/2} + C$ (d) None of these
[NDA (I) - 2012]
 6. Consider the following:
 I. $\int \log 10 dx = x + C$
 II. $\int 10^x dx = 10^x + C$
 Which of the above is/are correct?
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II
[NDA (I) - 2012]
 7. What is the value of $\int \frac{\ln x}{x} dx$?
 (a) $\frac{(\ln x)^2}{2} + C$ (b) $\frac{(\ln x)}{2} + C$
 (c) $(\ln x)^2 + C$ (d) None of these
[NDA (II) - 2012]
 8. What is the value of $\int a^x e^x dx$?
 (a) $\frac{a^x e^x}{\ln a} + C$ (b) $a^x e^x + C$
 (c) $\frac{a^x e^x}{\ln(ae)} + C$ (d) None of these
[NDA (II) - 2012]
 9. What is the value of $\int \left(\frac{1}{\cos^2 x} - \frac{1}{\sin^2 x} \right) dx$?
 (a) $2 \operatorname{cosec} 2x + C$ (b) $-2 \cot 2x + C$
 (c) $2 \sec 2x + C$ (d) $-2 \tan 2x + C$
[NDA (II) - 2012]
 10. What is $\int e^{\ln x} dx$ equal to?
 (a) $xe^{\ln x} + C$ (b) $-xe^{\ln x} + C$
 (c) $x + C$ (d) $\frac{x^2}{2} + C$
[NDA (I) - 2013]
 11. What is $\int \frac{dx}{x \ln x}$ equal to:
 (a) $\ln(\ln x) + C$ (b) $\ln x + C$
 (c) $(\ln x)^2 + C$ (d) None of these
[NDA (I) - 2013]
 12. What is $\int e^{e^x} e^x dx$ equal to?
 (a) $e^{e^x} + C$ (b) $2e^{e^x} + C$
 (c) $e^{e^x} e^x + C$ (d) $2e^{e^x} e^x + C$
[NDA (II) - 2013]
 13. What is $\int \sin^2 x dx + \int \cos^2 x dx$ equal to?
 (a) $x + C$ (b) $\frac{x^2}{2} + C$
 (c) $x^2 + C$ (d) None of these
[NDA (II) - 2013]
 14. What is $\int (x \cos x + \sin x) dx$ equal to?
 (a) $x \sin x + C$ (b) $x \cos x + C$
 (c) $-x \sin x + C$ (d) $-x \cos x + C$
[NDA (II) - 2013]
 15. What is $\int \frac{dx}{\sqrt{4+x^2}}$ equal to?
 (a) $\log \left| \sqrt{4+x^2} + x \right| + C$ (b) $\log \left| \sqrt{4+x^2} - x \right| + C$
 (c) $\sin^{-1} \left(\frac{x}{2} \right) + C$ (d) None of these
[NDA (II) - 2013]
- Directions (for next two):** Read the following information carefully and answer the questions given below:
 Consider the function $f'(x) = \sec^4 x + 4$ with $f(0) = 0$ and $f'(0) = 0$
16. What is $f'(x)$ equal to?
 (a) $\tan x - \frac{\tan^3 x}{3} + 4x$ (b) $\tan x + \frac{\tan^3 x}{3} + 4x$
 (c) $\tan x + \frac{\sec^3 x}{3} + 4x$ (d) $-\tan x - \frac{\tan^3 x}{3} + 4x$
[NDA (II) - 2014]
 17. What is $f(x)$ equal to?
 (a) $\frac{2 \ln \sec x}{3} + \frac{\tan^2 x}{6} + 2x^2$
 (b) $\frac{3 \ln \sec x}{2} + \frac{\cot^2 x}{6} + 2x^2$

(c) $\frac{4 \ln \sec x}{3} + \frac{\sec^2}{6} + 2x^2$

(d) $\ln \sec x + \frac{\tan^4 x}{12} + 2x^2$

[NDA (II) - 2014]

Directions (for next two): Read the following information carefully and answer these questions given below.

Consider $\int x \tan^{-1} x dx = A (x^2 + 1) \tan^{-1} x + Bx + C$ where, C is the constant of integration.

18. What is the value of A?

- (a) 1 (b) 1/2
(c) -1/2 (d) 1/4

[NDA (II) - 2014]

19. What is the value of B?

- (a) 1 (b) 1/2
(c) -1/2 (d) 1/4

[NDA (II) - 2014]

20. What is $\int \frac{dx}{\sqrt{x^2 + a^2}}$ equal to:

- (a) $\ln \left| \frac{x + \sqrt{x^2 + a^2}}{a} \right| + c$ (b) $\ln \left| \frac{x - \sqrt{x^2 + a^2}}{a} \right| + c$
(c) $\ln \left| \frac{x^2 + \sqrt{x^2 + a^2}}{a} \right| + c$ (d) None of these

[NDA (I) - 2015]

Directions (for next two): The integral $\int \frac{dx}{a \cos x + b \sin x}$ is of the form $\frac{1}{r} \ln \left[\tan \left(\frac{x + \alpha}{2} \right) \right]$.

21. What is r equal to?

- (a) $a^2 + b^2$ (b) $\sqrt{a^2 + b^2}$
(c) $a + b$ (d) $\sqrt{a^2 - b^2}$

[NDA (I) - 2015]

22. What is α equal to?

- (a) $\tan^{-1} \left(\frac{a}{b} \right)$ (b) $\tan^{-1} \left(\frac{b}{a} \right)$
(c) $\tan^{-1} \left(\frac{a+b}{a-b} \right)$ (d) $\tan^{-1} \left(\frac{a-b}{a+b} \right)$

[NDA (I) - 2015]

23. What is $\int \frac{x e^x dx}{(x+1)^2}$ equal to?

- (a) $(x+1)^2 e^x + C$ (b) $(x+1)e^x + C$
(c) $\frac{e^x}{x+1} + C$ (d) $\frac{e^x}{(x+1)^2} + C$

Where, C is the constant of integration.

[NDA (I) - 2015]

24. $\int \frac{dx}{1 + e^{-x}}$ is equal to:

- (a) $1 + e^{-x} + c$ (b) $\ln(1 + e^{-x}) + c$
(c) $\ln(1 + e^x) + c$ (d) $2 \ln(1 + e^{-x}) + c$

Where c is the constant of integration.

[NDA (II) - 2015]

Directions (for next two):

Consider $f'(x) = \frac{x^2}{2} - kx + 1$ such that $f(0) = 0$ and $f(3) =$

15.

25. The value of k is:

- (a) 5/3 (b) 3/5
(c) -5/3 (d) -3/5

[NDA (II) - 2015]

26. $f'' \left(-\frac{2}{3} \right)$ is equal to:

- (a) -1 (b) 1/3
(c) 1/2 (d) 1

[NDA (II) - 2015]

27. Let $f(x)$ and $g(x)$ be twice differentiable functions on $[0, 2]$ satisfying $f''(x) = g'(x)$, $f'(1) = 4$, $g'(1) = 6$, $f(2) = 3$ and $g(2) = 9$. Then what is $f(x) - g(x)$ at $x = 4$ equal to?

- (a) -10 (b) -6
(c) -4 (d) 2

[NDA (I) - 2016]

28. What is $\int \frac{x^4 - 1}{x^2 \sqrt{x^4 + x^2 + 1}} dx$ equal to?

- (a) $\sqrt{\frac{x^4 + x^2 + 1}{x}} + c$ (b) $\sqrt{x^4 + 2 - \frac{1}{x^2}} + c$
(c) $\sqrt{x^2 + \frac{1}{x^2} + 1} + c$ (d) $\sqrt{\frac{x^4 - x^2 + 1}{x}} + c$

[NDA (II) - 2016]

29. What is $\int e^{\sin x} \frac{x \cos^3 x - \sin x}{\cos^2 x} dx$ equal to?

- (a) $(x + \sec x)e^{\sin x} + c$ (b) $(x - \sec x)e^{\sin x} + c$
(c) $(x + \tan x)e^{\sin x} + c$ (d) $(x - \tan x)e^{\sin x} + c$

[NDA (II) - 2016]

30. What is $\int \frac{(x^{e-1} + e^{x-1})}{x^e + e^x} dx$ equal to?

- (a) $\frac{x^2}{2} + c$ (b) $\ln(x + e) + c$
(c) $\ln(x^e + e^x) + c$ (d) $\frac{1}{e} \ln(x^e + e^x) + c$

[NDA (I) - 2017]

31. Let $f(x)$ be an indefinite integral of $\sin^2 x$. Consider the following statements:

Statements

1. The function $f(x)$ satisfies $f(x + \pi) = f(x)$ for all real x .
2. $\sin^2(x + \pi) = \sin^2 x$ for all real x .

Which one of the following is correct in respect of the above statements?

- (a) Both the statements are true and statement 2 is the correct explanation of statement 1
(b) Both the statements are true but statement 2 is not the correct explanation of statement 1
(c) Statement 1 is true but statement 2 is false
(d) Statement 1 is false but statement 2 is true

[NDA (I) - 2017]

32. What is $\int \frac{dx}{x(x^7 + 1)}$ equal to?

- (a) $\frac{1}{2} \ln \left| \frac{x^7 - 1}{x^7 + 1} \right| + c$ (b) $\frac{1}{7} \ln \left| \frac{x^7 + 1}{x^7} \right| + c$

(c) $\ln \left| \frac{x^7 - 1}{7x} \right| + c$ (d) $\frac{1}{7} \ln \left| \frac{x^7}{x^7 + 1} \right| + c$

[NDA (I) - 2017]

33. Consider the following statements;
1. dy/dx at a point on the curve gives slope of the tangent at the point.
2. If $a(t)$ denotes acceleration of a particle, then $\int a(t) dt + c$ gives velocity of the particle.
3. If $s(t)$ gives displacement of a particle at time t , then ds/dt gives its acceleration at that instant.

Which of the above statements is/are correct?

- (a) 1 and 2 (b) Only 2
(c) Only 1 (d) 1, 2 and 3

[NDA (II) - 2017]

34. What is $\int \tan^{-1}(\sec x + \tan x) dx$ equal to?

- (a) $\frac{\pi x}{4} + \frac{x^2}{4} + c$ (b) $\frac{\pi x}{2} + \frac{x^2}{4} + c$
(c) $\frac{\pi x}{4} + \frac{\pi x^2}{4} + c$ (d) $\frac{\pi x}{4} - \frac{x^2}{4} + c$

[NDA (II) - 2017]

35. $\int (\ln x)^{-1} dx - \int (\ln x)^{-2} dx$ is equal to:

- (a) $x (\ln x)^{-1} + c$ (b) $x (\ln x)^{-2} + c$
(c) $x (\ln x) + c$ (d) $x (\ln x)^2 + c$

[NDA (II) - 2017]

36. $I_1 = \frac{d}{dx}(e^{\sin x})$

$I_2 = \lim_{h \rightarrow 0} \frac{e^{\sin(x+h)} - e^{\sin x}}{h}$

$I_3 = \int e^{\sin x} \cos x dx$, then which one of the following is correct?

- (a) $I_1 \neq I_2$ (b) $\frac{d}{dx}(I_3) = I_2$
(c) $\int I_3 dx = I_2$ (d) $I_2 = I_3$

[NDA (II) - 2017]

37. What is $\int \frac{dx}{2^x - 1}$ equal to?

- (a) $\ell n(2^x - 1) + C$ (b) $\frac{\ell n(1 - 2^{-x})}{\ell n 2} + C$
(c) $\frac{\ell n(2^{-x} - 1)}{2\ell n 2} + C$ (d) $\frac{\ell n(1 + 2^{-x})}{\ell n 2} + C$

[NDA (I) - 2018]

38. What is $\int \frac{dx}{a^2 \sin^2 x + b^2 \cos^2 x}$ equal to?

- (a) $c + \frac{1}{ab} \tan^{-1} \left(\frac{a \tan x}{b} \right)$
(b) $c - \frac{1}{ab} \tan^{-1} \left(\frac{b \tan x}{a} \right)$
(c) $c + \frac{1}{ab} \tan^{-1} \left(\frac{b \tan x}{a} \right)$
(d) None of the above

[NDA (II) - 2018]

39. What is $\int e^{\ln(\tan x)} dx$ equal to?

- (a) $\ln |\tan x| + c$ (b) $\ln |\sec x| + c$
(c) $\tan x + c$ (d) $e^{\tan x} + c$

Where c is the constant of integration.

[NDA (II) - 2018]

40. What is $\int \sin^3 x \cos x dx$ equal to?

- (a) $\cos^4 x + c$ (b) $\sin^4 x + c$
(c) $\frac{(1 - \sin^2 x)^2}{4} + c$ (d) $\frac{(1 - \cos^2 x)^2}{4} + c$

[NDA (II) - 2018]

41. What is $\int \ln(x)^2 dx$ equal to?

- (a) $2x \cdot \ln(x) - 2x + c$ (b) $\frac{2}{x} + c$
(c) $2x \cdot \ln(x) + c$ (d) $\frac{2 \ln(x)}{x} - 2x + c$

[NDA (I) - 2019]

42. What is $\int e^{x \ln(a)} dx$ equal to?

- (a) $\frac{a^x}{\ln(a)} + c$ (b) $\frac{e^x}{\ln(a)} + c$
(c) $\frac{e^x}{\ln(ae)} + c$ (d) $\frac{ae^x}{\ln(a)} + c$

[NDA (I) - 2019]

43. What is $\int \frac{dx}{2x^2 - 2x + 1}$ equal to ?

- (a) $\frac{\tan^{-1}(2x - 1)}{2} + c$ (b) $2 \tan^{-1}(2x - 1) + c$
(c) $\frac{\tan^{-1}(2x + 1)}{2} + c$ (d) $\tan^{-1}(2x - 1) + c$

[NDA (II) - 2019]

44. What is $\int \frac{dx}{x(1 + \ln x)^n}$ equal to ? ($n \neq 1$)

- (a) $\frac{1}{(n-1)(1 + \ln x)^{n-1}} + c$
(b) $\frac{(n-1)}{(1 + \ln x)^{n-1}} + c$
(c) $\frac{(n+1)}{(1 + \ln x)^{n+1}} + c$
(d) $-\frac{1}{(n-1)(1 + \ln x)^{n-1}} + c$

[NDA (II) - 2019]

45. If $p(x) = (4e)^{2x}$, then what is $\int p(x) dx$ equal to ?

- (a) $\frac{p(x)}{1 + 2 \ln 2} + c$ (b) $\frac{p(x)}{2(1 + 2 \ln 2)} + c$
(c) $\frac{2p(x)}{1 + \ln 4} + c$ (d) $\frac{p(x)}{1 + \ln 2} + c$

[NDA 2020]

46. What is $\int (e^{\log x} + \sin x) \cos x dx$ equal to ?

- (a) $\sin x + x \cos x + \frac{\sin^2 x}{2} + c$
(b) $\sin x - x \cos x + \frac{\sin^2 x}{2} + c$
(c) $x \sin x + \cos x + \frac{\sin^2 x}{2} + c$

(d) $x \sin x - \cos x + \frac{\sin^2 x}{2} + c$

[NDA 2020]

47. What is $\int \frac{dx}{x(x^n+1)}$ equal to ?

- (a) $\frac{1}{n} \ln \left(\frac{x^n}{x^n+1} \right) + c$ (b) $\ln \left(\frac{x^n+1}{x^n} \right) + c$
 (c) $\ln \left(\frac{x^n}{x^n+1} \right) + c$ (d) $\frac{1}{n} \ln \left(\frac{x^n+1}{x^n} \right) + c$

[NDA 2020]

48. What is the value of k such that integration of $\frac{3x^2+8-4k}{x}$ with respect to x, may be a rational function?

- (a) 0 (b) 1
 (c) 2 (d) -2

[NDA 2020]

49. What is $\int (e^{2 \log x + \log x^2}) dx$ equal to

- (a) $\frac{x^4}{4} + c$ (b) $\frac{x^3}{3} + c$
 (c) $\frac{2x^5}{5} + c$ (d) $\frac{x^5}{5} + c$

[NDA (I) - 2021]

50. What is $\int \frac{dx}{\sec x + \tan x}$ equal to ?

- (a) $\ln(\sec x) + \ln|\sec x + \tan x| + c$
 (b) $\ln(\sec x) - \ln|\sec x + \tan x| + c$
 (c) $\sec x \cdot \tan x - \ln|\sec x - \tan x| + c$
 (d) $\ln|\sec x + \tan x| - \ln(\sec x) + c$

[NDA (I) - 2021]

51. What is $\int \frac{dx}{\sec^2(\tan^{-1} x)}$ equal to ?

- (a) $\sin^{-1} x + c$ (b) $\tan^{-1} x + c$
 (c) $\sec^{-1} x + c$ (d) $\cos^{-1} x + c$

[NDA (I) - 2021]

52. What is $\int \frac{dx}{x(x^2+1)}$ equal to?

- (a) $\frac{1}{2} \ln \left(\frac{x^2}{x^2+1} \right) + c$ (b) $\ln \left(\frac{x^2}{x^2+1} \right) + c$
 (c) $\frac{3}{2} \ln \left(\frac{x^2}{x^2+1} \right) + c$ (d) $\frac{1}{2} \ln \left(\frac{x^2+1}{x^2} \right) + c$

[NDA (II) - 2021]

53. What is the integral of $f(x)=1+x^2+x^4$ with respect to x^2 ?

- (a) $x + \frac{x^3}{3} + \frac{x^5}{5} + c$
 (b) $\frac{x^3}{3} + \frac{x^5}{5} + c$
 (c) $x^2 + \frac{x^4}{4} + \frac{x^6}{6} + c$

(d) $x^2 + \frac{x^4}{2} + \frac{x^6}{3} + c$

[NDA (II) - 2021]

54. If $\int \sqrt{1-\sin 2x} dx = A \sin x + B \cos x + C$, where $0 < x < \frac{\pi}{4}$,

then which one of the following is correct?
 (a) $A + B = 0$ (b) $A + B - 2 = 0$
 (c) $A + B + 2 = 0$ (d) $A + B - 1 = 0$

[NDA (II) - 2021]

55. What is $\int (\sin x)^{-1/2} (\cos x)^{-3/2} dx$ equal to?

- (a) $\sqrt{\tan x} + c$ (b) $2\sqrt{\tan x} + c$
 (c) $\sqrt{\cot x} + c$ (d) $\sqrt{2 \tan x} + c$

[NDA (I) - 2022]

56. If $I_1 = \int \frac{e^x dx}{e^x + e^{-x}}$ and $I_2 = \int \frac{dx}{e^{2x} + 1}$, then what is $I_1 + I_2$ equal to?

- (a) $\frac{x}{2} + c$ (b) $x + c$
 (c) $\ln(e^x + e^{-x}) + c$ (d) $\ln(e^x - e^{-x}) + c$

[NDA (I) - 2022]

57. What is $\int (x^x)^2 (1 + \ln x) dx$ equal to?

- (a) $x^{2x} + c$ (b) $\frac{1}{2} x^{2x} + c$
 (c) $2x^{2x} + c$ (d) $\frac{1}{2} x^x + c$

[NDA 2022 (II)]

58. What is $\int e^x \{1 + \ln x + x \ln x\} dx$ equal to?

- (a) $x e^x \ln x + c$ (b) $x^2 e^x \ln x + c$
 (c) $x + e^x \ln x + c$ (d) $x e^x + \ln x + c$

[NDA 2022 (II)]

59. What is $\int \frac{(\cos x)^{1.5} - (\sin x)^{1.5}}{\sqrt{\sin x \cdot \cos x}} dx$ equal to?

- (a) $\sqrt{\sin x} - \sqrt{\cos x} + c$
 (b) $\sqrt{\sin x} + \sqrt{\cos x} + c$
 (c) $2\sqrt{\sin x} + 2\sqrt{\cos x} + c$
 (d) $\frac{1}{2} \sqrt{\sin x} + \frac{1}{2} \sqrt{\cos x} + c$

[NDA 2022 (II)]

Consider the following for the next two (02) items that follow:

Let $\int \frac{dx}{\sqrt{x+1} - \sqrt{x-1}} = \alpha(x+1)^{\frac{3}{2}} + \beta(x-1)^{\frac{3}{2}} + c$

60. What is the value of α ?

- (a) $\frac{1}{3}$ (b) $\frac{2}{3}$
 (c) 1 (d) $\frac{4}{3}$

[NDA-2024 (1)]

61. What is the value of β ?

- (a) $-\frac{2}{3}$ (b) $-\frac{1}{3}$
 (c) $\frac{1}{3}$ (d) $\frac{2}{3}$

[NDA-2024 (1)]

Consider the following for the next two (02) items that follow:

Given that $\int \frac{3\cos x + 4\sin x}{2\cos x + 5\sin x} dx =$

$\frac{\alpha x}{29} + \frac{\beta}{29} \ln |2\cos x + 5\sin x| + c$

62. What is the value of α ?

- (a) 7 (b) 13
(c) 17 (d) 26

[NDA-2024 (1)]

63. What is the value of β ?

- (a) 7 (b) 13
(c) 17 (d) 26

[NDA-2024 (1)]

Direction: Consider the following for the **two items** given below

Let $2 \int \frac{x^2 - 1}{\sqrt{x^2 + 1}} dx = U(x)V(x) - 3 \ln\{U(x) + V(x)\} + c$

64. What is $|U^2(x) - V^2(x)|$ equal to ?

- (a) 0 (b) 1
(c) 2 (d) 3

[NDA-2024 (2)]

65. What is $U(x)V(x)$ equal to ?

- (a) $\sqrt{x^2 + x^4}$ (b) $\sqrt{x + x^3}$
(c) $\frac{\sqrt{x^2 + x^4}}{2}$ (d) $2\sqrt{x^2 + x^4}$

[NDA-2024 (2)]

Consider the following for the one (01) items that follow:

Let the function $y = (1 - \cos x)^{-1}$, where $x \neq 2n\pi$ and n is an integer.

66. What is $\int y dx$ equal to?

- (a) $-\tan(x/2) + c$ (b) $-\cot(x/2) + c$
(c) $\tan(x/2) + c$ (d) $\cot(x/2) + c$

where c is the constant of integration

[NDA-2025 (1)]

For the following next **two** items:

Let $u = \int e^x \cos x dx$ and $v = \int e^x \sin x dx$

67. What is $u + v$ equal to ?

- (a) $-\frac{du}{dx}$ (b) $-\frac{dv}{dx}$
(c) $\frac{du}{dx}$ (d) $\frac{dv}{dx}$

[NDA-2025 (2)]

68. Consider the following

I. $\frac{du}{dx} = -v$ II. $\frac{dv}{dx} = -u$

Which of the statements given above is/are correct?

- (a) I only (b) II only
(c) Both I and II (d) Neither I nor II

[NDA-2025 (2)]

69. What is $\int \frac{dx}{10^x + 10^{-x}}$ equal to?

- (a) $\tan^{-1}(10^x) + c$ (b) $(\ln 10) \tan^{-1}(10^x) + c$
(c) $\frac{1}{\ln 10} \tan^{-1}(10^x) + c$ (d) $\tan^{-1}(10^x + 10^{-x}) + c$

[NDA-2025 (2)]

ANSWER KEY

1.	d	2.	b	3.	a	4.	b	5.	c	6.	a	7.	a	8.	c	9.	a	10.	d
11.	a	12.	a	13.	a	14.	a	15.	a	16.	b	17.	a	18.	b	19.	c	20.	a
21.	b	22.	a	23.	c	24.	c	25.	c	26.	d	27.	a	28.	c	29.	b	30.	d
31.	d	32.	d	33.	a	34.	a	35.	a	36.	b	37.	b	38.	a	39.	b	40.	d
41.	a	42.	a	43.	d	44.	d	45.	b	46.	c	47.	a	48.	c	49.	d	50.	d
51.	b	52.	a	53.	d	54.	b	55.	b	56.	b	57.	b	58.	a	59.	c	60.	a
61.	c	62.	d	63.	a	64.	b	65.	a	66.	b	67.	d	68.	d	69.	c		

Solutions

Sol. 1. (d)

$$\int \frac{\sin \sqrt{x}}{\sqrt{x}} dx = 2 \int \frac{\sin \sqrt{x}}{2\sqrt{x}} dx$$

$$\begin{aligned} \text{let } \sqrt{x} &= t \\ \Rightarrow 2 \int \sin t dt &= -2 \cos t + c \\ &= -2 \cos \sqrt{x} + c \end{aligned}$$

Sol. 2. (b)

$$\int e^x \left(\sqrt{x} + \frac{1}{2\sqrt{x}} \right) dx = e^x \sqrt{x} + c$$

Sol. 3. (a)

$$\begin{aligned} \int \sin^{-1}(\cos x) dx &= \int \pi/2 - \cos^{-1}(\cos x) dx \\ &= \int \left(\frac{\pi}{2} - x \right) dx = \frac{\pi x}{2} - \frac{x^2}{2} + c \end{aligned}$$

Sol. 4. (b)

$$\begin{aligned} \int \left(\frac{1}{\sin^2 x \cos^2 x} \right) dx &= \int \left(\frac{\sin^2 x + \cos^2 x}{\sin^2 x \cos^2 x} \right) dx \\ \int \left(\frac{1}{\sin^2 x} + \frac{1}{\cos^2 x} \right) dx &= \int (\csc^2 x + \sec^2 x) dx \\ &= \tan x - \cot x + c \end{aligned}$$

Sol. 5. (c)

$$\begin{aligned} &= \int (x^2 + 1)^{5/2} x dx \\ \text{put } x^2 + 1 &= t \text{ then } 2x dx = dt \\ &= (1/2) \int t^{5/2} dt \\ &= 1/7 (t)^{7/2} + C \\ &= 1/7 (x^2 + 1)^{7/2} + C \end{aligned}$$

Sol. 6. (a)

$$\begin{aligned} \text{I. } \int \log_{10} dx &= \int 1 dx = x + C \\ \text{II. } \int 10^x dx &= 10^x \log_e 10 + C \end{aligned}$$

Sol. 7. (a)

$$\begin{aligned} \int \frac{\ln x}{x} dx \\ \text{put } \ln x = t \text{ then } (1/x) dx = dt \\ = \int t dt = \frac{t^2}{2} + c = \frac{(\ln x)^2}{2} + c \end{aligned}$$

Sol. 8. (c)

$$\begin{aligned} \int a^x e^x dx &= \int (ae)^x dx \\ &= \frac{(ae)^x}{\ln ae} + c = \frac{a^x e^x}{\ln ae} + c \end{aligned}$$

Sol. 9. (a)

$$\begin{aligned} \int \left(\frac{1}{\cos^2 x} - \frac{1}{\sin^2 x} \right) dx \\ = \int (\sec^2 x - \csc^2 x) dx \\ = \tan x + \cot x + c = \frac{\sin x}{\cos x} + \frac{\cos x}{\sin x} + c \end{aligned}$$

$$\frac{\sin^2 x + \cos^2 x}{\sin x \cos x} + c = \frac{2}{2 \sin x \cos x} + c = 2 \csc 2x + c$$

Sol. 10. (d)

$$\int e^{\ln x} dx = \int x dx = \frac{x^2}{2} + c$$

Sol. 11. (a)

$$\begin{aligned} \int \frac{dx}{x \ln x} \\ \text{put } \ln x = t \text{ then } (1/x) dx = dt \end{aligned}$$

$$\int \frac{dt}{t} = \ln t + c = \ln(\ln x) + c$$

Sol. 12. (a)

$$\begin{aligned} \int e^{e^x} e^x dx &= \\ \text{put } t &= e^{e^x} \text{ then } e^{e^x} e^x dx = dt \\ \Rightarrow \int dt &= t + c = e^{e^x} + c \end{aligned}$$

Sol. 13. (a)

$$\int \sin^2 x dx + \int \cos^2 x dx = \int (\sin^2 x + \cos^2 x) x dx = \int 1 \cdot dx = x + c$$

Sol. 14. (a)

$$\begin{aligned} \int (x \cos x + \sin x) dx \\ = x \sin x - \int \sin x dx + \int \sin x dx \\ = x \sin x + c \end{aligned}$$

Sol. 15. (a)

$$\begin{aligned} \therefore \int \frac{1}{\sqrt{a^2 + x^2}} dx &= \log \left| x + \sqrt{a^2 + x^2} \right| + c \\ \therefore \int \frac{1}{\sqrt{4 + x^2}} dx &= \log \left| x + \sqrt{4 + x^2} \right| + c \end{aligned}$$

Sol. 16. (b)

$$\begin{aligned} f''(x) &= \sec^4 x + 4 \\ f'(x) &= \int (\sec^4 x + 4) dx \\ &= \int (\sec^2 x \cdot \sec^2 x + 4) dx \\ &= \int ((1 + \tan^2 x) \sec^2 x + 4) dx \\ f'(x) &= \tan x + \frac{\tan^3}{3} + 4x + C \end{aligned}$$

$$\begin{aligned} \text{put } x = 0 \text{ and } f'(0) &= 0 \\ \text{then } c &= 0 \end{aligned}$$

$$f'(x) = \tan x + \frac{\tan^3}{3} + 4x$$

$$\text{with } f(0) = 0$$

Sol. 17. (a)

$$\begin{aligned} f''(x) &= \tan x + \frac{\tan^3}{3} + 4x \\ f(x) &= \int \left(\tan x + \frac{\tan^2 x \cdot \tan x}{3} + 4x \right) dx \\ &= \int \left(\tan x + \frac{(\sec^2 x - 1) \tan x}{3} + 4x \right) dx \\ &= \int \left(\frac{2}{3} \tan x + \frac{(\sec^2 x) \tan x}{3} + 4x \right) dx \\ &= \frac{2}{3} \ln \sec x + \frac{\tan^2 x}{6} + 2x^2 + k \end{aligned}$$

$$\text{put } x = 0 \text{ and } f(0) = 0 \text{ then } k = 0$$

Sol. 18. (b)

$$\begin{aligned} \int x \tan^{-1} x dx \\ = \tan^{-1} x \left(\frac{x^2}{2} \right) - \int \frac{1}{1+x^2} \cdot \frac{x^2}{2} dx \\ = \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} \int \frac{(x^2 + 1 - 1) dx}{1+x^2} \\ = \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} \int \left(1 - \frac{1}{1+x^2} \right) dx \end{aligned}$$

$$= \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} x + \frac{1}{2} \tan^{-1} x + c$$

$$= \left(\frac{x^2 + 1}{2} \right) \tan^{-1} x - \frac{1}{2} x + c$$

$$A = \frac{1}{2}$$

Sol. 19. (c)

$$B = -1/2$$

Sol. 20. (a)

$$\therefore \int \frac{1}{\sqrt{a^2 + x^2}} dx = \log \left| \frac{x + \sqrt{a^2 + x^2}}{a} \right| + c$$

Sol. 21. (b)

$$\begin{aligned} \int \frac{dx}{a \cos x + b \sin x} \\ \text{put } a = r \sin \alpha \text{ and } b = r \cos \alpha \\ \frac{1}{r} \int \frac{dx}{\sin \alpha \cos x + \cos \alpha \sin x} \\ = \frac{1}{r} \int \frac{dx}{\sin(x + \alpha)} = \frac{1}{r} \int \csc(x + \alpha) dx \\ = 1/r \ln |\csc(x + \alpha) - \cot(x + \alpha)| + c \\ = \frac{1}{r} \ln \left| \tan \left(\frac{x + \alpha}{2} \right) \right| + c \\ = r = \sqrt{a^2 + b^2} \end{aligned}$$

Sol. 22. (a)

$$\alpha = \tan^{-1}(a/b)$$

Sol. 23. (c)

$$\begin{aligned} \int \frac{x e^x dx}{(x+1)^2} &= \int e^x \left(\frac{x}{(x+1)^2} \right) dx \\ &= \int e^x \left(\frac{x+1-1}{(x+1)^2} \right) dx \\ &= \int e^x \left(\frac{1}{x+1} - \frac{1}{(x+1)^2} \right) dx \\ &= \int e^x \left(\frac{1}{x+1} \right) + c \end{aligned}$$

Sol. 24. (c)

$$\int \frac{dx}{1+e^{-x}} = \int \frac{e^x dx}{1+e^x} = \log(1+e^x) + c$$

Sol. 25. (c)

$$\begin{aligned} f'(x) &= \frac{x^2}{2} - kx + 1 \\ f(x) &= \int \left(\frac{x^2}{2} - kx + 1 \right) dx \\ f(x) &= \frac{x^3}{6} - k \frac{x^2}{2} + x + c \\ f(0) &= 0, \text{ then } \Rightarrow c = 0 \\ f(3) &= 15 \\ f(x) &= \frac{x^3}{6} - k \frac{x^2}{2} + x \end{aligned}$$

$$15 = \frac{27}{6} - k \frac{9}{2} + 3 \Rightarrow k = -5/3$$

Sol. 26. (d)

$$f'(x) = \frac{x^2}{2} - kx + 1$$

$$f''(x) = x - k$$

$$f''\left(-\frac{2}{3}\right) = -\frac{2}{3} - \left(-\frac{5}{3}\right) = 1$$

Sol. 27. (a)

$$f''(x) = g''(x),$$

$$f'(x) = g'(x) + c$$

$$\text{put } x = 1$$

$$4 = 6 + c \text{ then } c = -2$$

$$f(x) = g(x) - 2x + d$$

$$\text{put } x = 2$$

$$3 = 9 - 2(2) + d \text{ then } d = -2$$

$$\text{at } x = 4$$

$$f(4) - g(4) = -2(-4) - 2 = -10$$

Sol. 28. (c)

$$\int \frac{x^4 - 1}{x^2 \sqrt{x^4 + x^2 + 1}} dx = \int \frac{x - \frac{1}{x^3}}{\sqrt{x^2 + 1 + \frac{1}{x^2}}} dx$$

$$\text{Let } \Rightarrow x^2 + 1 + \frac{1}{x^2} = t$$

$$\Rightarrow 2x - \frac{2}{x^3} dx = dt$$

$$\int \frac{1}{\sqrt{t}} dt = \sqrt{t} + c$$

$$= \sqrt{x^2 + 1 + \frac{1}{x^2}} + c$$

Sol. 29. (b)

$$\int e^{\sin x} (x \cos x - \tan x \sec x) dx$$

$$= \int (e^{\sin x} x \cos x - e^{\sin x} \tan x \sec x) dx$$

$$= \int (e^{\sin x} x \cos x dx + e^{\sin x})$$

$$- e^{\sin x} (\tan x \sec x - \sec x e^{\sin x} \cos x) dx$$

$$= \int d(x e^{\sin x}) - \int d(\sec x e^{\sin x})$$

$$= x e^{\sin x} - \sec x e^{\sin x} + c$$

$$= (x - \sec x) e^{\sin x} + c$$

Sol. 30. (d)

$$\int \frac{(x^{e-1} + e^{x-1})}{x^e + e^x} dx = \frac{1}{e} \int \frac{(e x^{e-1} + e^x)}{x^e + e^x} dx$$

$$= 1/e \log(x^e + e^x) + c$$

Sol. 31. (d)

$$\int \sin^2 x dx = \int \left(\frac{1 - \cos 2x}{2} \right) dx$$

$$= \frac{x}{2} - \frac{\sin 2x}{4} + c$$

this function is not periodic

Sol. 32. (d)

$$\int \frac{dx}{x(x^7 + 1)} = \int \frac{x^6 dx}{x^7(x^7 + 1)}$$

$$= \frac{1}{7} \int \frac{dt}{(t+1)} = \frac{1}{7} \log \left| \frac{t}{t+1} \right| + c$$

$$= \frac{1}{7} \log \left| \frac{x^7}{x^7 + 1} \right| + c$$

Sol. 33. (a)

$\frac{ds}{dt}$ gives velocity so 3rd statement is correct.

Sol. 34. (a)

$$\int \tan^{-1}(\sec x + \tan x) dx$$

$$= \int \tan^{-1} \left(\frac{1 + \sin x}{\cos x} \right) dx$$

$$= \int \tan^{-1} \left(\frac{\left(\sin \frac{x}{2} + \cos \frac{x}{2} \right)^2}{\cos^2 \frac{x}{2} - \sin^2 \frac{x}{2}} \right) dx = \int \tan^{-1} \left(\frac{\cos \frac{x}{2} + \sin \frac{x}{2}}{\cos \frac{x}{2} - \sin \frac{x}{2}} \right) dx$$

$$= \int \tan^{-1} \left(\frac{1 + \tan \frac{x}{2}}{1 - \tan \frac{x}{2}} \right) dx = \int \tan^{-1} \left(\tan \left(\frac{\pi}{4} + \frac{x}{2} \right) \right) dx$$

$$= \int \frac{\pi}{4} + \frac{x}{2} dx = \frac{\pi x}{4} + \frac{x^2}{4} + c$$

Sol. 35. (a)

$$\int (\ln x)^{-1} dx - \int (\ln x)^{-2} dx$$

$$\text{put } \ln x = t, \text{ then } x = e^t \text{ and } dx = e^t dt$$

$$\int e^t [(t)^{-1} - (t)^{-2}] dx = \int e^t \left(\frac{1}{t} - \frac{1}{t^2} \right) dx$$

$$= e^t \cdot \frac{1}{t} + c = \frac{x}{\ln x} + c$$

Sol. 36. (b)

$$I_1 = \int e^{\sin x} \cos x dx$$

$$I_2 = \int e^{\sin x} \cos x dx$$

$$I_3 = \int e^{\sin x} \cos x dx$$

Sol. 37. (b)

$$\int \frac{2^{-x} dx}{1 - 2^{-x}} = \frac{1}{\ln a} \int \frac{2^{-x} \ln a dx}{1 - 2^{-x}} = \frac{\log(1 - 2^{-x})}{\ln a} + c$$

Sol. 38. (a)

$$\int \frac{dx}{a^2 \sin^2 x + b^2 \cos^2 x} = \int \frac{\sec^2 x dx}{a^2 \tan^2 x + b^2}$$

$$= \frac{1}{a^2} \int \frac{\sec^2 x dx}{\tan^2 x + \frac{b^2}{a^2}} = \frac{1}{a^2} \int \frac{dt}{t^2 + \left(\frac{b}{a} \right)^2}$$

$$= \frac{1}{a^2} \cdot \frac{1}{\frac{b}{a}} \tan^{-1} \left(\frac{t}{\frac{b}{a}} \right) + c = \frac{1}{ab} \tan^{-1} \left(\frac{a \tan x}{b} \right) + c$$

Sol. 39. (b)

$$\int e^{\ln(\tan x)} dx = \int \tan x dx = \ln |\sec x| + c$$

Sol. 40. (d)

$$\int \sin^3 x \cos x dx$$

$$\text{put } \sin x = t, \text{ then } \cos x dx = dt$$

$$\int t^3 dx = \frac{t^4}{4} + c = \frac{\sin^4 x}{4} + c$$

$$= \frac{(1 - \cos^2 x)^2}{4} + c$$

Sol. 41. (a)

$$\int \ln(x)^2 dx = \int 2 \ln(x) dx = 2x \ln(x) - 2x + c$$

Sol. 42. (a)

$$\int e^{x \ln(a)} dx = \int e^{\ln(a)^x} dx = \int a^x dx = \frac{a^x}{\ln a} + c$$

Sol. 43. (d)

$$\int \frac{dx}{2x^2 - 2x + 1} = \frac{1}{2} \int \frac{dx}{x^2 - x + \frac{1}{2}}$$

$$= \frac{1}{2} \int \frac{dx}{x^2 - x + \frac{1}{4} + \frac{1}{4}} + \frac{1}{2} \int \frac{dx}{\left(x - \frac{1}{2} \right)^2 + \frac{1}{4}}$$

$$= \tan^{-1} \left(\frac{x - \frac{1}{2}}{\frac{1}{2}} \right) + c = \tan^{-1}(2x - 1) + c$$

Sol. 44. (d)

$$\text{put } 1 + \log x = t$$

$$= \int \frac{dt}{t^n} = \frac{t^{-n+1}}{-n+1} = \frac{t^{-(n-1)}}{1-n}$$

$$= -\frac{1}{(n-1)(1+\ln x)^{n-1}} + c$$

Sol. 45. (b)

$$\int (4e)^{2x} dx = \frac{(4e)^{2x}}{2 \ln(4e)} + c$$

$$= \frac{(4e)^{2x}}{2[\ln(4) + \ln(e)]} + c = \frac{p(x)}{2(1 + 2 \ln 2)} + c$$

Sol. 46. (c)

$$\int (e^{\log x} + \sin x) \cos x dx$$

$$= \int (x + \sin x) \cos x dx$$

$$= \int (x \cdot \cos x + \sin x \cdot \cos x) dx$$

$$x \sin x - \int \sin x dx + \int \frac{1}{2} \sin 2x dx$$

$$x \sin x + \cos x - \frac{1}{2} \cos 2x + c$$

$$x \sin x + \cos x - \frac{1}{2} (1 - 2 \sin^2 x) + c$$

$$x \sin x + \cos x - \frac{1}{4} + \frac{\sin^2 x}{2} + c$$

$$x \sin x + \cos x + \frac{\sin^2 x}{2} + c$$

Sol. 47. (a)

$$\int \frac{dx}{x(x^n + 1)} = \int \frac{x^{n-1} dx}{x^n(x^n + 1)}$$

$$\text{put } x^n = t$$

$$\frac{1}{n} \int \frac{dt}{t(t+1)} = \frac{1}{n} \int \left(\frac{1}{t} - \frac{1}{t+1} \right) dt$$

$$= \frac{1}{n} \log \left(\frac{t}{t+1} \right) = \frac{1}{n} \log \left(\frac{x^n}{x^n + 1} \right)$$

Sol. 48. (c)

$$\frac{3x^2 + 8 - 4k}{x} = 3x + \frac{8 - 4k}{x}$$

Integration of $1/x = \log x$ (not a rational value)

$$\text{so } 8 - 4k = 0$$

$$k = 2$$

Sol. 49. (d)

$$\int (e^{2 \log x + \log x^2}) dx$$

$$= \int (e^{\log x^2 + \log x^2}) dx = \int (e^{2 \log x^2}) dx$$

$$= \int (e^{\log x^4}) dx = \int x^4 dx = \frac{x^5}{5} + c$$

Sol. 50. (d)

$$\int \frac{dx}{\sec x + \tan x}$$

$$= \int (\sec x - \tan x) dx$$

$$= \ln |\sec x + \tan x| - \ln |\sec x| + c$$

Sol. 51. (b)

$$\int \frac{dx}{\sec^2(\tan^{-1} x)}$$

$$= \int \frac{dx}{\sec^2(\sec^{-1} \sqrt{1+x^2})}$$

$$= \int \frac{dx}{[\sec(\sec^{-1} \sqrt{1+x^2})]^2} = \int \frac{dx}{1+x^2}$$

$$= \tan^{-1}x + c$$

Sol. 52. (a)

$$\int \frac{dx}{x(x^2+1)} = \int \frac{x dx}{x(x^2+1)}$$

$$\text{put } x^n = t$$

$$\frac{1}{2} \int \frac{dt}{t(t+1)} = \frac{1}{2} \int \left(\frac{1}{t} - \frac{1}{t+1} \right) dt$$

$$= \frac{1}{2} \log \left(\frac{t}{t+1} \right) = \frac{1}{2} \log \left(\frac{x^2}{x^2+1} \right)$$

Sol. 53. (d)

$$f(x) = 1 + x^2 + x^4 \text{ w.r.t } x^2$$

$$x^2 = y \quad f(y) = 1 + y + y^2$$

$$\int f(y) dy = y + \frac{y^2}{2} + \frac{y^3}{3} + C$$

$$= x^2 + \frac{x^4}{2} + \frac{x^6}{3} + C$$

Sol. 54. (b)

$$\sqrt{1 - \sin 2x} = \cos x - \sin x \text{ in } (0, \pi/4)$$

$$\int (\cos x - \sin x) dx = \sin x + \cos x + c$$

$$A = 1 \text{ and } B = 1$$

$$\text{Then } A + B = 2$$

$$A + B - 2 = 0$$

Sol. 55. (b)

$$\int (\sin x)^{-1/2} (\cos x)^{-3/2} dx$$

$$= \int \frac{1}{\sqrt{\sin x} \sqrt{\cos x} \cos x} dx$$

$$= \int \frac{1}{\sqrt{\tan x} \cos^2 x} dx$$

$$= \int \frac{\sec^2 x}{\sqrt{\tan x}} dx$$

$$\text{Put } \tan x = t$$

$$\sec^2 x dx = dt$$

$$= \int \frac{1}{\sqrt{t}} dt$$

$$= 2\sqrt{t} + c$$

$$= \sqrt{\tan x} + c$$

Sol. 56. (b)

$$I_1 = \int \frac{e^x dx}{e^x + e^{-x}}$$

$$I_2 = \int \frac{dx}{e^{2x} + 1} = \int \frac{e^{-x}}{e^x + e^{-x}} dx$$

$$I_1 + I_2 = \int \frac{e^x}{e^x + e^{-x}} dx + \int \frac{e^{-x}}{e^x + e^{-x}} dx$$

$$= \int \frac{e^x + e^{-x}}{e^x + e^{-x}} dx$$

$$= \int 1 dx = x + c$$

Sol. 57. (b)

$$\int (x^x)^2 (1+\log x) dx$$

$$= \int (x^x) (x^x) (1+\log x) dx$$

$$\text{Put } x^x = t$$

$$(x^x) (1+\log x) dx = dt$$

$$\Rightarrow \int t dt = t^2/2 + c$$

$$= \frac{(x^x)^2}{2} + c$$

$$= \frac{x^{2x}}{2} + c$$

Sol. 58. (a)

$$\int e^x (1+\log x + x \log x) dx$$

$$\because \int e^x (f(x) + f'(x)) dx = e^x f(x) + c$$

$$\text{Here } \frac{d}{dx} (x \log x) = 1 + \log x$$

$$\text{So } \int e^x (1 + \log x + x \log x) dx = e^x \cdot x$$

$$\log x + c$$

Sol. 59. (c)

$$\Rightarrow \int \frac{\cos x}{\sqrt{\sin x}} - \frac{\sin x}{\sqrt{\cos x}} dx$$

$$\text{By substitution in first integration put } \sin x = t$$

$$\text{and in 2nd integration put } \cos x = t$$

$$\text{Then answer is}$$

$$\Rightarrow 2\sqrt{\sin x} + 2\sqrt{\cos x} + c$$

Sol. 60. (a)

$$\frac{1}{\sqrt{x+1} - \sqrt{x-1}} \times \frac{\sqrt{x+1} + \sqrt{x-1}}{\sqrt{x+1} + \sqrt{x-1}}$$

$$= \frac{1}{2} \times \sqrt{x+1} + \sqrt{x-1}$$

$$= \frac{1}{2} \times \int \sqrt{x+1} + \sqrt{x-1} dx$$

$$= \frac{1}{2} \left[\frac{(x+1)^{3/2}}{3/2} + \frac{(x-1)^{3/2}}{3/2} \right]$$

$$= \frac{1}{3} \left[(x+1)^{3/2} + (x-1)^{3/2} \right]$$

$$\text{so } \alpha = 1/3$$

Sol. 61. (c)

$$\text{by above solution}$$

$$\beta = 1/3$$

Sol. 62. (d)

$$\int \frac{3 \cos x + 4 \sin x}{2 \cos x + 5 \sin x} dx$$

$$\text{In } \frac{a \cos x + b \sin x}{c \cos x + d \sin x} \text{ type of integration we will}$$

$$\text{write this expression as below}$$

$$\text{Numerator}$$

$$= A \frac{d}{dx} (\text{denominator}) + B (\text{denominator})$$

$$\text{so}$$

$$3 \cos x + 4 \sin x = A \frac{d}{dx} (2 \cos x + 5 \sin x) + B (2 \cos x + 5 \sin x)$$

$$3 \cos x + 4 \sin x = A(-2 \sin x + \cos x) + B(2 \cos x + 5 \sin x)$$

$$3 \cos x + 4 \sin x = (5A + 2B) \cos x + (-2A + 5B) \sin x$$

$$\text{by comparing coefficients of } \sin x \text{ and } \cos x \text{ we get}$$

$$3 = 5A + 2B$$

$$4 = -2A + 5B$$

$$\text{by solving equation } A = 7/29 \text{ and } B = 26/29$$

$$3 \cos x + 4 \sin x = \frac{7}{29} (2 \cos x + 5 \sin x) + \frac{26}{29} (2 \cos x + 5 \sin x)$$

$$\int \frac{3 \cos x + 4 \sin x}{2 \cos x + 5 \sin x} dx =$$

$$\frac{7}{29} \int \frac{-2 \sin x + 5 \cos x}{2 \cos x + 5 \sin x} dx + \frac{26}{29} \int \frac{2 \cos x + 5 \sin x}{2 \cos x + 5 \sin x} dx$$

$$= \frac{7}{29} \int \frac{-2 \sin x + 5 \cos x}{2 \cos x + 5 \sin x} dx + \frac{26}{29} \int 1 dx$$

$$= \frac{26}{29} x + \frac{7}{29} \log |2 \cos x + 5 \sin x + c|$$

$$\text{so } \alpha = 26$$

NOTE: this question can solve by differentiate right hand side than compare with left hand side given integration.

Sol. 63. (a)

$$\text{by above solution } \beta = 7$$

Sol. 64. (b)

$$2 \int \frac{x^2 - 1}{\sqrt{x^2 + 1}} dx = U(x)V(x) - 3 \ln \{U(x) + V(x)\} + c$$

$$2 \int \frac{x^2 - 1}{\sqrt{x^2 + 1}} dx = 2 \int \frac{x^2 + 1 - 2}{\sqrt{x^2 + 1}} dx$$

$$= 2 \int \left(\sqrt{x^2 + 1} + \frac{-2}{\sqrt{x^2 + 1}} \right) dx$$

$$= 2 \left[\frac{x}{2} \sqrt{x^2 + 1} + \frac{1}{2} \log (x + \sqrt{x^2 + 1}) - 2 \log (x + \sqrt{x^2 + 1}) \right]$$

$$= 2 \left[\frac{x}{2} \sqrt{x^2 + 1} - \frac{3}{2} \log (x + \sqrt{x^2 + 1}) \right] + c$$

$$= x \sqrt{x^2 + 1} - 3 \log (x + \sqrt{x^2 + 1}) + c$$

$$\text{compare with}$$

$$U(x)V(x) - 3 \log \{U(x) + V(x)\} + c$$

$$\text{so}$$

$$U(x) = x \text{ and } V(x) = \sqrt{x^2 + 1}$$

$$|U^2(x) - V^2(x)| = |x^2 - x^2 - 1| = 1$$

Sol. 65. (a)

$$\text{by above solution}$$

$$U(x) = x \text{ and } V(x) = \sqrt{x^2 + 1}$$

$$U(x).V(x) = x \sqrt{x^2 + 1} = \sqrt{x^4 + x^2}$$

Sol. 66. (b)

$$\int \frac{1}{1 - \cos x} dx = \int \frac{1}{2 \sin^2 \frac{x}{2}} dx$$

$$= \int \frac{1}{2} \sec^2 \frac{x}{2} dx = \frac{1}{2} \left(-\cot \frac{x}{2} \right) \frac{1}{1/2} = -\cot \frac{x}{2} + c$$

Sol. 67. (d)

$$u + v = \int e^x \cos x dx + \int e^x \sin x dx$$

$$= \int e^x (\sin x + \cos x) dx = e^x \sin x + c \quad \dots (i)$$

$$v = \int e^x \sin x dx$$

$$\text{Differentiate both side w.r.t. } x$$

$$\frac{dv}{dx} = e^x \sin x \quad \dots (ii)$$

$$\text{By (i) and (ii)}$$

$$u + v = \frac{dv}{dx}$$

Sol. 68. (d)

$$\frac{du}{dx} = e^x \cos x \neq -v$$

$$\frac{dv}{dx} = e^x \sin x \neq -u$$

$$\text{Both statements are wrong}$$

Sol. 69. (c)

$$\int \frac{dx}{10^x + 10^{-x}} = \int \frac{10^x dx}{10^{2x} + 1} = \frac{1}{\ln 10} \int \frac{10^x \ln 10 dx}{10^{2x} + 1}$$

$$\text{Let } 10^x = t$$

$$10^x \ln 10 dx = dt$$

$$= \frac{1}{\ln 10} \int \frac{dt}{t^2 + 1} = \frac{1}{\ln 10} \tan^{-1} (10^x) + c$$